



CAFFEINE & CADAVER
MBBS BIOCHEMISTRY NOTES SERIES

PYQ 2010–2025

MODULE 05

Nucleotide Chemistry & Metabolism

MBBS BIOCHEMISTRY · PAPER 1

UNIVERSITY	SUBJECT	EDITION
MBBS	Biochemistry	Last-Day Revision System

MASTER PYQ INDEX (2010–2025)

2011

2012

2013

2015-S

2016-S

2017

2018

2019

2021

2023

2025

 EXAM PRIORITY MAP

● **TIER 1 (High Yield – Must Master):** Purine & Pyrimidine De Novo Synthesis, Salvage Pathways, Gout & Hyperuricaemia, Allopurinol (Suicide Inhibition), Lesch-Nyhan Syndrome, ADA Deficiency, Nucleotide Analogs as Drugs.

● **TIER 2 (Medium Yield – Review):** Purine Catabolism & Uric Acid, CPS I vs CPS II, Orotic Aciduria (Type I vs Urea Cycle defect).

● **TIER 3 (Low Yield – Fast Read):** Nucleotide Structure (bonds, ribose vs deoxyribose), Essay Question Templates.

00: 🔥 Last Day Revision Mode

- **Focus on TIER 1:** These topics comprise over 80% of all exam questions. Study the active recall Q&As in Section 10.
- **Review the Exam Traps:** Pay special attention to the ⚠️ EXAM TRAP boxes injected in each section. They highlight common student mistakes.
- **Tier 3 sections are collapsed:** Standard HTML details/summary elements have been used to collapse low-yield details, saving you valuable time. Click to expand only if needed.

01 Nucleotide Structure — TIER 3

Expand TIER 3: Nucleotide Structure Section Details

⚡ A **Nucleotide** = Nitrogenous base + Pentose sugar + Phosphate group. Remove phosphate → **Nucleoside**. Remove sugar → just the base.

Purines (Bicyclic)

Pyrimidine fused to imidazole ring

- Adenine (A): amino group at C6
- Guanine (G): keto at C6, amino at C2

Pyrimidines (Monocyclic)

Single 6-membered ring

- Cytosine (C): DNA & RNA
- Thymine (T): DNA only (C5 methyl)
- Uracil (U): RNA only (no methyl)

Sugars (Pentose)

N-glycosidic bond (β-config)

- Ribose: in RNA (2'-OH present)
- Deoxyribose: in DNA (2'-H, no OH)

Key Bonds

Connecting components

- N-glycosidic: base → sugar (C1')
- Phosphoester: sugar C5' → phosphate
- Phosphodiester: links chain (3'–5')

COMPONENT	PURINE	PYRIMIDINE
RING STRUCTURE	Bicyclic (double ring)	Monocyclic (single ring)
MEMBERS	Adenine, Guanine	Cytosine, Thymine (DNA), Uracil (RNA)
WATSON-CRICK PAIRING	A pairs with T (2 H-bonds); G pairs with C (3 H-bonds)	Paired with purines
DE NOVO SYNTHESIS	Built ON pre-formed ribose-5-P scaffold	Ring built first, then attached to ribose
CATABOLISM END PRODUCT	Uric acid (humans)	CO ₂ + NH ₃ + β-alanine/β-aminoisobutyrate
SALVAGE PATHWAY	HGPRT (HG), APRT (A)	Not prominent in humans

MNEMONIC – BASES IN DNA VS RNA

DNA: "A TiGht Cell" | RNA: "A UGly Cell"

DNA: Adenine, Thymine, Guanine, Cytosine.

RNA: Adenine, Uracil, Guanine, Cytosine.

Key EQ: DNA is alkali resistant, RNA is alkali labile. RNA's 2'-OH group facilitates cyclic phosphodiester intermediate formation under alkaline conditions, leading to rapid autohydrolysis. DNA has 2'-H, rendering it stable.

02 Purine De Novo Synthesis ● TIER 1

SOURCES OF ATOMS IN THE PURINE RING

PURINE RING ATOM ORIGINS

Nitrogen atoms

- N1 ← Aspartate
- N3, N9 ← Glutamine
- N7 ← Glycine

Carbon atoms

- C4, C5 ← Glycine
- C2, C8 ← N¹⁰-formyl-THF
- C6 ← CO₂

MNEMONIC – PURINE RING SOURCES

"Glutamine Attacks Good Cells, Aspartate Fights CO₂"

Glycine (C4,C5,N7 - entire unit) · **G**lutamine (N3,N9 amide nitrogens) · N¹⁰-Formyl-THF (C2,C8) · **A**spartate (N1) · CO₂ (C6).



EXAM TRAP: Do not confuse the rate-limiting enzyme of Purine De Novo Synthesis (Glutamine-PRPP amidotransferase) with that of Pyrimidine synthesis (CPS II). Also, remember that purine ring atoms are built directly ON the ribose-5-phosphate scaffold, unlike pyrimidines where the ring is synthesized first and then attached to PRPP.

PURINE DE NOVO SYNTHESIS – KEY STEPS TO IMP

Ribose-5-phosphate
(From HMP Shunt/PPP)

↓ **PRPP Synthetase (+ ATP)**

PRPP (Activated Sugar Scaffold)
(5-phosphoribosyl-1-pyrophosphate)

↓ **Glutamine:PRPP amidotransferase**

RATE LIMITING STEP ⚡

5-Phosphoribosylamine (PRA)
First committed step · Feedback inhibited by AMP, GMP, IMP

↓ **9 Enzymatic Steps (Adds Gly, Formyl-THF, Asp, Gln, CO₂)**

IMP (Inosine Monophosphate)
Common branchpoint intermediate containing Hypoxanthine base

↓ **Reciprocal Pathway Branching**

AMP Branch

Requires GTP & Aspartate
Inhibited by AMP

GMP Branch

Requires ATP & Glutamine
Inhibited by GMP (IMP Dehydrogenase target)

Regulation of Purine De Novo Synthesis

- **PRPP Synthetase:** Allosterically inhibited by AMP, GMP, IMP (end-product regulation).
- **Glutamine:PRPP Amidotransferase (RLS):** Synergistically inhibited by AMP and GMP together. Activated by substrate PRPP.
- **Cross-Regulation:** AMP synthesis requires energy from GTP; GMP synthesis requires energy from ATP. This reciprocity ensures a balanced purine nucleotide pool for DNA/RNA synthesis.
- **Site of Synthesis:** Primarily the **liver**. Brain, RBCs, and polymorphonuclear leukocytes have low de novo capacity and rely heavily on salvage.

03 Pyrimidine De Novo Synthesis

● **TIER 1**



Key Distinction: In pyrimidine biosynthesis, the **six-membered ring is completely assembled FIRST** as Orotic Acid, and only then attached to the PRPP ribose sugar. In purine biosynthesis, the ring is built atom-by-atom directly onto PRPP.

ATOM SOURCES IN THE PYRIMIDINE RING

PYRIMIDINE RING ATOM ORIGINS

N1, C4, C5, C6

Aspartate

C2CO₂ (HCO₃⁻)**N3**

Glutamine (Amide N)

MNEMONIC – PYRIMIDINE RING SOURCES

"ACG builds the Pyrimidine Ring"

Aspartate (N1, C4, C5, C6) · **C**O₂ (C2) · **G**lutamine (N3). Much simpler than purine donors (only 3 components required).



EXAM TRAP: OTC deficiency (urea cycle defect) leads to cytosolic accumulation and spillover of carbamoyl phosphate. This fuels the cytosolic pyrimidine pathway, causing hyperuricosuria/orotic aciduria, but it does NOT cause megaloblastic anaemia. In contrast, UMP Synthase deficiency (true Orotic Aciduria) ALWAYS presents with megaloblastic anaemia.

PYRIMIDINE DE NOVO PATHWAY – KEY STEPS

Glutamine + CO₂ + 2 ATP↓ **Carbamoyl Phosphate Synthetase II (CPS II)**

RATE LIMITING ⚡

Carbamoyl Phosphate (Cytosolic)

Inhibited by UTP · Activated by PRPP

↓ **Aspartate Transcarbamoylase (ATCase) + Dihydroorotase****Dihydroorotate**

Formed by CAD trifunctional polypeptide in cytosol

↓ **Dihydroorotate Dehydrogenase (DHODH) (Mitochondrial Membrane)****Orotate (Orotic Acid)**

First fully formed pyrimidine ring

↓ **UMP Synthase (Bifunctional: OPRT + ODCase) + PRPP****UMP (Uridine Monophosphate)**

Parent pyrimidine nucleotide

↓ **Phosphorylation & Amination****UTP → CTP**

Via CTP Synthetase (uses Glutamine)

dUMP → dTMP

Via Thymidylate Synthase (uses Folate)

Clinical Core — Orotic Aciduria

- **Type I Orotic Aciduria:** Inherited deficiency of the bifunctional enzyme **UMP Synthase** (OPRT and/or ODCase). Orotic acid accumulates and crystallizes in urine.
- **Clinical Triad:** Megaloblastic anaemia (refractory to B12/folate, because pyrimidines are required for DNA synthesis/erythropoiesis), urinary orotic acid crystals, and physical/developmental delay.
- **Treatment:** **Uridine supplementation** (bypasses the block to provide UMP/dUTP, and its feed-forward conversion to UTP directly inhibits CPS II, halting overproduction of orotate).

COMPARISON – CPS I VS CPS II

FEATURE	CPS I	CPS II
SUBCELLULAR LOCATION	Mitochondria	Cytosol
METABOLIC PATHWAY	Urea Cycle (Nitrogen disposal)	Pyrimidine De Novo Synthesis
PRIMARY NITROGEN SOURCE	Free Ammonia (NH ₃)	Glutamine (Amide group)
ALLOSTERIC ACTIVATOR	N-Acetylglutamate (NAG) (obligate)	PRPP
FEEDBACK INHIBITOR	None	UTP (and UDP)
TISSUE DISTRIBUTION	Liver and intestinal mucosa only	All nucleated cells (highest in dividing cells)

04 Purine Salvage Pathways ● TIER 1

 **Salvage:** Recycles free purine bases (from cellular turnover or diet) back into active nucleotides. It consumes significantly less metabolic energy (ATP) than the 10-step de novo pathway.

 **EXAM TRAP:** In Lesch-Nyhan syndrome, the severe hyperuricaemia is not just due to a failure to recycle hypoxanthine and guanine. It is critically amplified because PRPP accumulates (since it's not used by HGPRT) and directly activates Glutamine-PRPP amidotransferase, driving de novo purine synthesis to go 10x faster!

PURINE SALVAGE OVERVIEW

Hypoxanthine + PRPP
↓ HGPRT enzyme
IMP

Guanine + PRPP
↓ HGPRT enzyme
GMP

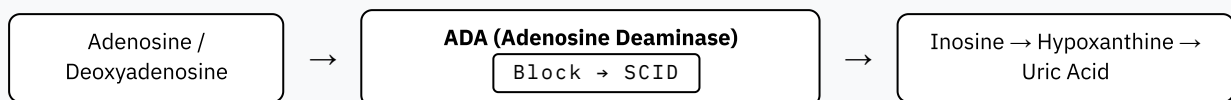
Adenine + PRPP
↓ APRT enzyme
AMP

Clinical Core — Lesch-Nyhan Syndrome (LNS)

- **Enzyme Defect:** Complete deficiency of **HGPRT** (Hypoxanthine-Guanine Phosphoribosyltransferase). X-linked recessive inheritance.
- **Pathophysiology:** Hypoxanthine and Guanine accumulate and are converted to Uric Acid. High levels of unused PRPP stimulate the rate-limiting enzyme of the de novo pathway, causing hyper-activation of purine synthesis and massive hyperuricaemia.
- **Clinical Triad:** Hyperuricaemia (tophi, gout, renal urate stones), severe neurological dysfunction (choreoathetosis, spastic cerebral palsy, intellectual disability), and **compulsive self-mutilation** (lip and finger biting).
- **Neurological Link:** The brain lacks the capacity for de novo purine synthesis and is completely dependent on HGPRT salvage. Loss of salvage causes depletion of GTP and damages dopaminergic pathways in the basal ganglia.

ADENOSINE DEAMINASE (ADA) DEFICIENCY

ADENOSINE CATABOLIC PATHWAY & BLOCK



SCID Biochemical Mechanism: ADA deficiency leads to accumulation of Deoxyadenosine, which cellular kinases phosphorylate into **dATP**. High dATP is a potent allosteric **inhibitor** of Ribonucleotide Reductase (RNR). RNR inhibition halts the synthesis of all other dNTPs (dCTP, dGTP, dTTP) → completely blocking S-phase DNA synthesis. Lymphocytes (T, B, NK cells) are extremely sensitive and undergo rapid apoptosis, causing **Severe Combined Immunodeficiency (SCID)**.

05 Purine Catabolism & Uric Acid — ● TIER 2

PURINE DEGRADATION PATHWAY TO URIC ACID

AMP / Adenosine

GMP / Guanosine

↓ Deamination & Ribose Cleavage

Hypoxanthine

Guanine

↓ Xanthine Oxidase (XO) / Guanase

Xanthine (Intermediate)

↓ Xanthine Oxidase (XO)

TARGET OF ALLOPURINOL ⚡

URIC ACID

End product in humans · Low solubility · Excreted via kidney & gut

EVOLUTIONARY LOSS OF URICASE

Most mammals possess the enzyme **uricase**, which converts uric acid into highly soluble, easily excretable **allantoin**. During hominid evolution, the gene encoding uricase was silenced by nonsense mutations. As a result, humans and higher primates excrete uric acid as the final product, which is susceptible to precipitation in tissues at elevated concentrations.

SPECIES	END PRODUCT OF PURINE CATABOLISM	SOLUBILITY PROPERTIES
HUMANS, GREAT APES, BIRDS	Uric Acid / Monosodium Urate	Relatively insoluble; risks gout/calculi
MOST OTHER MAMMALS	Allantoin	Highly soluble and non-toxic
MARINE INVERTEBRATES	Free Ammonia (NH ₃)	Water-soluble and easily diluted

06 Gout & Hyperuricaemia ● TIER 1



Gout: Inflammatory arthritis triggered by the deposition of needle-shaped **monosodium urate (MSU) crystals** in joints and soft tissues (forming tophi) due to chronic hyperuricaemia (Serum urate > 7 mg/dL).



EXAM TRAP: Allopurinol is a suicide inhibitor, but it is NOT the active inhibitor itself. Allopurinol is first oxidized by Xanthine Oxidase to **Oxipurinol (Alloxanthine)**, which then binds irreversibly to the reduced Mo⁴⁺ active site. This is a classic question.

ETIOLOGY OF HYPERURICAEMIA

Overproduction (10% of cases)

- HGPRT deficiency (LNS)
- PRPP Synthetase overactivity
- Rapid cell turnover (leukaemia, tumor lysis)
- Von Gierke's (G6P accumulation redirects to PPP)
- Diet rich in purines (red meat, shellfish)

Underexcretion (90% of cases)

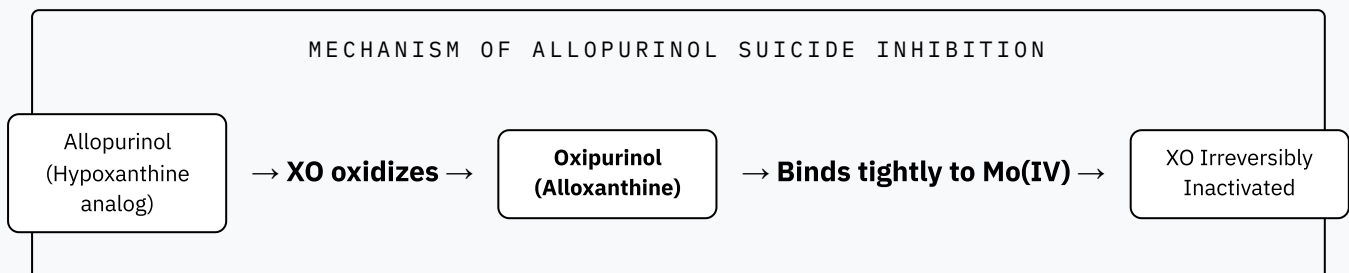
- Chronic Kidney Disease (CKD)
- Thiazide & Loop Diuretics
- Low-dose Aspirin (blocks secretory transporters)
- Ketoacidosis or Lactic Acidosis (competition for proximal tubule secretion)

Biochemical Link — Alcohol & Gout

- **Competiton:** Ethanol metabolism elevates NADH, converting pyruvate to lactate. Lactate competitively inhibits the renal urate transporter (URAT1/OAT), reducing uric acid excretion.
- **ATP Depletion:** Rapid metabolism of ethanol to acetate consumes ATP, generating ADP and AMP. AMP is catabolized to uric acid, increasing the urate pool.
- **Precursors:** Yeast-rich alcoholic drinks like beer contain high amounts of highly absorbable guanosine.

THERAPEUTIC PHARMACOLOGY OF GOUT

MECHANISM OF ALLOPURINOL SUICIDE INHIBITION



DRUG	TARGET & MECHANISM	CLINICAL USE
ALLOPURINOL	Suicide/mechanism-based inhibitor of Xanthine Oxidase	First-line chronic gout management & Tumor Lysis prophylaxis
FEBUXOSTAT	Non-purine, selective competitive inhibitor of XO	Alternative chronic gout therapy (if renal impairment exists)
COLCHICINE	Binds tubulin to inhibit neutrophil microtubule polymerization	First-line abortive agent for acute gouty attacks
PROBENECID	Inhibits proximal tubule URAT1 reabsorption of urate	Uricosuric agent (increases renal clearance of uric acid)
RASBURICASE	Recombinant fungal Uricase (converts urate to allantoin)	Rapid treatment of pediatric Tumor Lysis Syndrome

07 Lesch-Nyhan & ADA Deficiency ● TIER 1

CORE COMPARISON – HIGH-YIELD GENETIC DISORDERS

FEATURE	LESCH-NYHAN SYNDROME	ADA DEFICIENCY SCID
DEFICIENT ENZYME	HGPRT (Hypoxanthine-Guanine Phosphoribosyltransferase)	Adenosine Deaminase (ADA)
METABOLIC SCAPE	Purine Salvage Pathway	Purine Nucleoside Degradation
ACCUMULATED METABOLITE	Uric Acid (serum & urine) & substrate PRPP	dATP (deoxyadenosine triphosphate) in lymphocytes
PRIMARY PATHOLOGY	Basal ganglia dopaminergic deficit & renal lithiasis	Ribonucleotide Reductase inhibition leading to cell death
CLINICAL PHENOTYPE	Choreoathetosis, spasticity, compulsive self-mutilation, gout	Recurrent opportunistic infections, failure to thrive (SCID)
INHERITANCE	X-linked Recessive (affects males)	Autosomal Recessive
THERAPEUTICS	Allopurinol (reduces gout only; no neurological cure)	Enzyme replacement (PEG-ADA), Gene Therapy, BMT

08 Nucleotide Analogs as Drugs ● TIER 1

 **Mechanism:** Synthetic nucleotide and nucleoside analogs exploit cellular nucleotide metabolism enzymes to incorporate into DNA/RNA (causing chain termination or mutations) or directly inhibit critical DNA replication enzymes in cancer or viruses.

 **EXAM TRAP:** 5-Fluorouracil (5-FU) and Methotrexate (MTX) both inhibit thymidine synthesis, but 5-FU is a suicide inhibitor of **Thymidylate Synthase** (directly blocking dTMP synthesis), whereas Methotrexate is a competitive inhibitor of **Dihydrofolate Reductase (DHFR)**, blocking the regeneration of THF.

CLINICALLY SIGNIFICANT NUCLEOTIDE / NUCLEOSIDE ANALOGS

DRUG NAME	STRUCTURAL ANALOG OF	TARGET ENZYME	MECHANISM OF ACTION	CLINICAL USE
6-MERCAPTOPURINE (6-MP)	Hypoxanthine	HGPRT (activation) → PRPP Amidotransferase	Converts to thio-IMP → feedback inhibits de novo purine synthesis	Acute Lymphoblastic Leukemia (ALL), Crohn's Disease
5-FLUOROURACIL (5-FU)	Uracil	Thymidylate Synthase (TS)	Forms covalent ternary complex with TS and THF (Suicide Inhibition)	Colorectal cancer, pancreatic cancer, breast cancer
METHOTREXATE (MTX)	Folic Acid	Dihydrofolate Reductase (DHFR)	Competitive inhibitor; depletes tetrahydrofolate (THF) needed for purine/TMP	Cancer, Rheumatoid Arthritis, Psoriasis
ZIDOVUDINE (AZT)	Thymidine	Viral Reverse Transcriptase	Lacks a 3'-OH group; acts as an obligate DNA chain terminator	HIV-1 treatment and prevention of mother-to-child transmission
ACYCLOVIR	Guanosine	Viral Thymidine Kinase (activation) → DNA Pol	Phosphorylated selectively by viral TK → incorporates and terminates chain	Herpes Simplex Virus (HSV) & Varicella Zoster Virus (VZV)
CYTARABINE (ARA-C)	Cytidine (with arabinose sugar)	DNA Polymerase	Incorporates into replication fork; sterically hinders DNA elongation	Acute Myeloid Leukemia (AML) & Non-Hodgkin Lymphoma
MYCOPHENOLATE MOFETIL	Purine Intermediate	IMP Dehydrogenase (IMPDH)	Reversibly inhibits conversion of IMP to GMP; selectively depletes GMP in T/B cells	Immunosuppressive for renal/heart transplantation
HYDROXYUREA	Urea Analog	Ribonucleotide Reductase (RNR)	Scavenges free radical on RNR active site, blocking ribonucleotide to dNTP conversion	Myeloproliferative disorders, Sickle Cell Disease (induces HbF)

09 Orotic Aciduria & CPS I vs II — ● TIER 2

THE BIFURCATION OF CARBAMOYL PHOSPHATE USE

Mitochondrial CPS I

$\text{NH}_3 + \text{CO}_2 + 2 \text{ATP}$
Activated by N-acetylglutamate
↓
Carbamoyl Phosphate (Mito)
↓ OTC enzyme
Citruilline → **Urea Cycle**

Cytosolic CPS II

$\text{Glutamine} + \text{CO}_2 + 2 \text{ATP}$
Activated by PRPP · Inhibited by UTP
↓
Carbamoyl Phosphate (Cyto)
↓ ATCase enzyme
Orotic Acid → UMP → **Pyrimidine Pathway**

DIFFERENTIATING URINARY OROTIC ACID ELEVATION

When orotic acid is elevated, check for hyperammonaemia and anemia to locate the defect.

FEATURE	UMP SYNTHASE DEFICIENCY (OROTIC ACIDURIA TYPE I)	OTC DEFICIENCY (MITOCHONDRIAL UREA CYCLE DEFECT)
SUBCELLULAR LOCATION	Cytosolic Pyrimidine Pathway	Mitochondrial Urea Cycle
AMMONIA LEVELS	Normal (no defect in urea excretion)	Severely Elevated (Hyperammonaemia)
HAEMATOLOGICAL STATUS	Megaloblastic Anaemia (refractory to B12/folate)	Normal red blood cells
THERAPEUTICS	Exogenous Uridine	Protein restriction + Nitrogen scavengers (Sodium Benzoate)

PURINE METABOLISM & SALVAGE VIVAS

PYQ 2013, 2015-S What are the sources of N and C atoms in the purine ring? Draw and label.

Nitrogen Donors:

N1: Aspartate (amino nitrogen).

N3, N9: Glutamine (amide nitrogen).

N7: Glycine.

Carbon Donors:

C4, C5: Glycine (entire molecule).

C2, C8: N¹⁰-Formyl-tetrahydrofolate (one-carbon units).

C6: CO₂ (bicarbonate).

Assembly note: The purine ring is constructed directly on a pre-formed **ribose-5-phosphate** scaffold. The first committed step is the synthesis of 5-phosphoribosylamine by **Glutamine:PRPP amidotransferase** (the rate-limiting step). It takes 10 enzymatic reactions to close the imidazole and pyrimidine rings, forming IMP.

PYQ 2018, 2025 What is the purine salvage pathway? Explain its clinical significance.

Definition: The salvage pathway is a metabolic system that recycles free purine bases (Hypoxanthine, Guanine, Adenine) released from nucleic acid degradation back into active nucleotides (IMP, GMP, AMP).

Enzymatic Reactions:

HGPRT: Hypoxanthine + PRPP → IMP + PPi | Guanine + PRPP → GMP + PPi

APRT: Adenine + PRPP → AMP + PPi

Clinical & Physiological Significance:

- Energy Conservation:** Avoids the high ATP cost of the 10-step de novo synthesis pathway.
- Brain Homeostasis:** The brain has exceptionally low levels of de novo synthesis enzymes and is completely dependent on salvage to maintain its GTP/ATP pools. Deficiencies present with severe central nervous system lesions.
- Lesch-Nyhan Syndrome:** Inherited absence of HGPRT causes uric acid overproduction, self-mutilation, choreoathetosis, and gout.
- Active Pharmacology:** Immunosuppressants and antineoplastic drugs (e.g., 6-MP) require conversion by HGPRT to their nucleotide forms to exert their therapeutic effects.

PYQ 2012, 2019 How does allopurinol lower uric acid? Why is it called a suicide inhibitor?

Mechanism of Action:

- Allopurinol is a structural isomer of hypoxanthine.
- It binds to the enzyme **Xanthine Oxidase (XO)**, which oxidizes it into **Oxipurinol (Alloxanthine)**.
- Oxipurinol binds tightly and non-covalently to the reduced molybdenum ion (Mo⁴⁺) in the XO active site.
- This traps the enzyme in an inactive state, preventing the oxidation of hypoxanthine and xanthine into uric acid.

Suicide Inhibition Definition: Suicide inhibition (mechanism-based inhibition) occurs when an enzyme converts a structural analogue substrate into a reactive, high-affinity inhibitor that permanently binds and inactivates the same enzyme's active site. Because XO's own catalytic action produces the inactive complex, allopurinol is classified as a suicide inhibitor.

Outcome: Serum urate levels decline, and highly soluble hypoxanthine and xanthine accumulate and are excreted in urine.

Core Comparison Differentiate between Purine and Pyrimidine de novo synthesis.

FEATURE	PURINE DE NOVO	PYRIMIDINE DE NOVO
RING ASSEMBLY	Built step-by-step directly ON the ribose-5-P scaffold	Ring is built first (Orotate), then attached to PRPP
RATE-LIMITING STEP	Glutamine:PRPP amidotransferase	Carbamoyl Phosphate Synthetase II (CPS II)
ATOM DONORS	Glycine, Glutamine, Aspartate, THF, CO ₂	Aspartate, Glutamine, CO ₂
METABOLIC SITE	Cytosol of liver primarily	Cytosol and Mitochondrial Inner Membrane (DHODH)
PRIMARY PRODUCT	IMP (Inosine Monophosphate)	UMP (Uridine Monophosphate)

PYQ 2011-S Why does ADA deficiency selectively destroy T and B lymphocytes?**Selectivity Mechanism:**

1. Lymphocytes have exceptionally high activity of **Deoxynucleoside Kinase** (which phosphorylates deoxyadenosine to dAMP) and low activity of **5'-nucleotidase** (which degrades dAMP).
2. Consequently, when Adenosine Deaminase is absent, deoxyadenosine is rapidly converted to **dATP** within lymphocytes.
3. dATP is a potent allosteric inhibitor of **Ribonucleotide Reductase (RNR)**, blocking all DNA synthesis.
4. Other tissue cells have high 5'-nucleotidase activity and bypass this block, clearing the deoxyadenosine before toxic dATP can accumulate. Lymphocytes, lacking this pathway, die via apoptosis.

Advance Topic Contrast UMP Synthase Deficiency with OTC Deficiency.

Both diseases present with high orotic acid in urine. They are differentiated by checking blood ammonia and blood cell indices:

OTC Deficiency: Mitochondrial urea cycle block. Carbamoyl phosphate overflows into the cytosol, driving pyrimidine synthesis and elevating orotate. Presents with **severe hyperammonaemia** (lethargy, coma) but **normal hematocrit**.

UMP Synthase Deficiency (Orotic Aciduria): Cytosolic pyrimidine block. Presents with **normal blood ammonia** but **profound Megaloblastic Anaemia** due to nucleotide starvation in erythroid progenitors. Treated with oral **Uridine**.

11 Essay Question Templates — TIER 3

Expand TIER 3: Essay Question Templates Details

PYQ 2012-S, 2019 Explain why Allopurinol is used in the treatment of chronic Gout.

Standard Exam Answer Draft:

- 1. Target:** Chronic Gout is caused by hyperuricaemia, resulting in Monosodium Urate crystal precipitation in joints.
- 2. Therapeutic Goal:** Lower serum uric acid below 6.0 mg/dL to dissolve existing crystals and prevent new deposits.
- 3. Mechanism:** Allopurinol is a structural isomer of hypoxanthine. Xanthine Oxidase (XO) converts it to Oxipurinol (Alloxanthine).
- 4. Inhibition Class:** Oxipurinol binds tightly and irreversibly to the Mo^{4+} active site of XO, acting as a ****suicide inhibitor****.
- 5. Metabolic Shift:** Conversion of hypoxanthine and xanthine to uric acid is blocked. Soluble hypoxanthine and xanthine accumulate and are excreted, while serum uric acid levels drop.
- 6. De Novo Downregulation:** Clearance of PRPP by salvage pathways reduces substrates for de novo purine synthesis, further lowering urate levels.

PYQ 2012-S, 2016-S, 2018-S, 2021 Explain the biochemical rationale for alcohol-induced gouty attacks.

Standard Exam Answer Draft:

- 1. Metabolic Block:** Oxidation of ethanol by Alcohol Dehydrogenase generates NADH. High NADH/NAD⁺ favors the reduction of pyruvate to lactate.
- 2. Excretion Competition:** Lactate accumulates and competitively inhibits the organic anion transporter (URAT1) in renal proximal tubule cells, preventing renal uric acid secretion and elevating serum urate.
- 3. Purine Load:** Yeast in fermented drinks (such as beer) contains high levels of guanosine, which is catabolized directly into uric acid.
- 4. ATP Turnover:** High acetate production in liver cells drives ATP conversion to AMP. Accumulation of AMP feeds into the purine catabolic pathway, accelerating urate synthesis.

PYQ 2011-S Trace the biochemical steps from ADA deficiency to SCID.

Standard Exam Answer Draft:

- 1. Metabolic Defect:** Absence of Adenosine Deaminase prevents the conversion of adenosine to inosine, and deoxyadenosine to deoxyinosine.
- 2. Metabolite Accumulation:** Deoxyadenosine accumulates and is phosphorylated by lymphoid deoxycytidine kinase to form toxic ****dATP****.
- 3. Enzyme Inhibition:** High dATP binds to the activity site of ****Ribonucleotide Reductase (RNR)****, shutting down the conversion of all ribonucleotides to deoxyribonucleotides.
- 4. DNA Synthesis Arrest:** DNA replication is arrested in T, B, and NK cells, inducing apoptosis. This results in the profound lymphopenia characteristic of Severe Combined Immunodeficiency (SCID).

🔥 EXAM DAY HIGH-YIELD QUICKFIRE

Rate-limiting enzyme of Purine De Novo synthesis

Glutamine:PRPP amidotransferase. Inhibited by AMP, GMP, IMP. Activated by PRPP.

Rate-limiting enzyme of Pyrimidine De Novo synthesis

Carbamoyl Phosphate Synthetase II (CPS II) in the cytosol. Inhibited by UTP. Activated by PRPP.

Active metabolite of Allopurinol

Oxipurinol (Alloxanthine), which suicide-inhibits Xanthine Oxidase at its Mo^{4+} site.

Target enzyme and mechanism of 5-Fluorouracil (5-FU)

Thymidylate Synthase. Acts as a suicide inhibitor to deplete cellular dTMP and cause cell death.

Mechanism of Methotrexate (MTX)

Competitive inhibitor of Dihydrofolate Reductase (DHFR). Blocks folic acid reduction, depleting THF for purine and dTMP synthesis.

Primary diagnostic marker for OTC vs UMP synthase deficiency

Blood Ammonia: Severely elevated in OTC deficiency; normal in UMP synthase deficiency (Orotic Aciduria).
Megaloblastic anemia is only present in Orotic Aciduria.

Why does the brain rely on salvage pathways?

The brain has low expression of Glutamine:PRPP amidotransferase, making de novo synthesis impossible. It relies on HGPRT salvage.

MODULE 05 MASTER MNEMONICS

"PURE As Gold" & "CUT the Py"

- **Purines:** Adenine & Guanine (**PURE As Gold**). Bicyclic structure.
- **Pyrimidines:** Cytosine, Uracil, Thymine (**CUT the Py**). Monocyclic structure.
- **Lesch-Nyhan Syndrome:** **G**out, **H**yperuricaemia, **P**ast spasticity/choreoathetosis, **R**etardation, **T**enacious self-mutilation (corresponds to **HGPRT**).

MBBS Biochemistry - Module 05: Nucleotide Chemistry & Metabolism

Compiled from PYQ 2010-2025

*Crafted with precision to illuminate the pathways of biochemistry
and inspire your journey in medicine.*

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